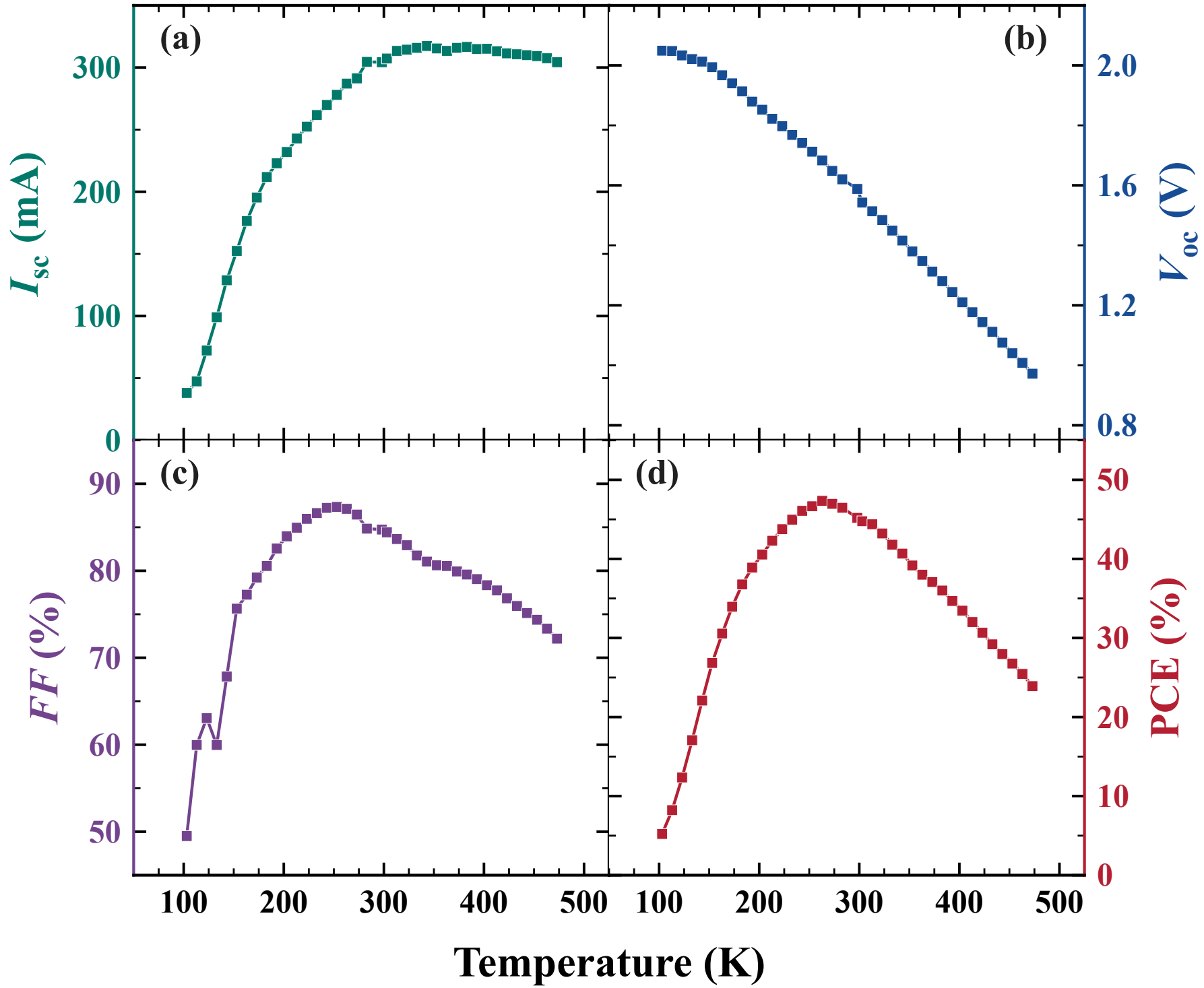
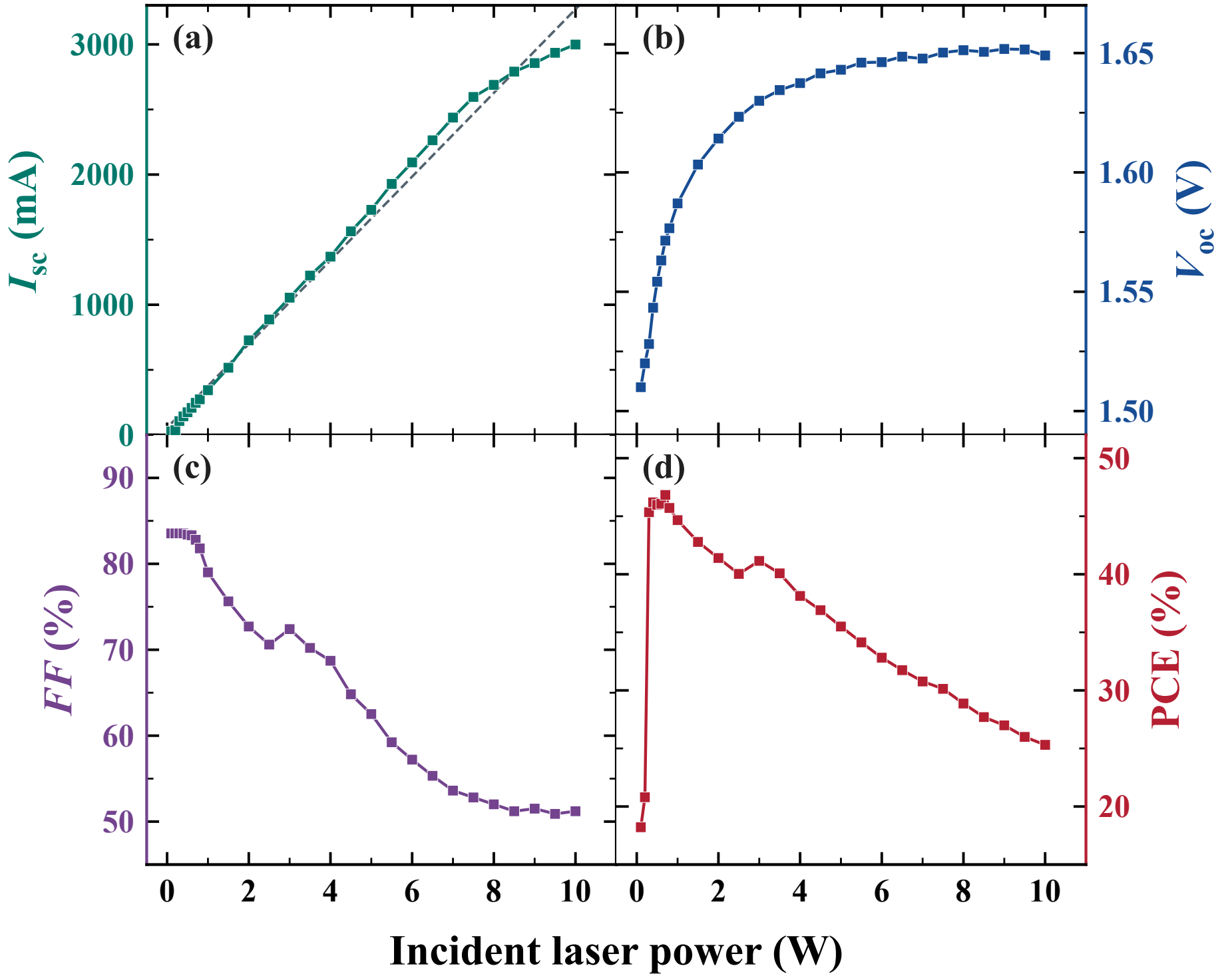

(a)
(b)
(c)
(d)
I_{sc} (mA)
V_{oc} (V)
FF (%)
PCE (%)
Temperature (K)
0
100
200
300
0.8
1.2
1.6
2.0
50
60
70
80
90
0
10
20
30
40
50
100
200
300
400
500


(a)
(b)
(c)
(d)
I_{sc} (mA)
V_{oc} (V)
FF (%)
PCE (%)
Incident laser power (W)
0
1000
2000
3000
1.50
1.55
1.60
1.65
50
60
70
80
90
20
30
40
50
0
2
4
6
8
10